Using Big Data and Hadoop in the Student Learning Process - Enhancing the Educational Process through Real Experience

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***Abstract*— In recent years, with the development of IoT technologies, digitization in all spheres of life, the generation of data of any nature in huge quantities, the organization, storage, processing and analysis of big data is becoming more and more relevant. Hadoop as an ecosystem provides very powerful tools for working with big data. The environment itself is free, and with the availability of appropriate hardware for its installation, familiarization with such frameworks is very useful for students. This report presents the advantages and disadvantages of using Hadoop in student education and how the use of these modern environments can improve the educational process, expand the practical skills of students and prepare them for their future professional challenges in the field of processing and analysis of big data.**

Keywords—big data, Hadoop, Hive, Hue, Education

I. Introduction

In today's digital age, Big Data plays a key role in business and research. As the volume of data grows, the need for efficient methods of storage, processing and analysis becomes more relevant than ever. One of the most popular and powerful technologies for managing big data is Apache Hadoop.

In the modern educational process, big data plays an increasingly significant role. Processing and analyzing large volumes of data can provide valuable guidance and knowledge that enriches students' experiences and prepares them for the challenges of the future work environment. In this context, Apache Hadoop and its related ecosystem provide powerful tools for working with big data that can be integrated into the curriculum. This article examines how the use of big data and Hadoop can enhance the educational process and prepare students for the future.

II. Big Data and Its Importance

Big data is characterized by volume, velocity and variety (the so-called 3V), and recently Value and Veracity (5V) have been added [1]. What makes big data different and more special is not just its characteristics, but rather the inability of traditional data management and processing methods to cope with it. This data comes from various sources such as social networks, sensor devices, Internet of Things (IoT) and others.

The amount of data has grown tremendously, and this leads to the need to look for solutions with the possibility of better scalability. Existing and widely used relational (ACID) databases cannot handle this problem. The goal of the new solutions is to satisfy the BASE paradigm (basically available, soft state, eventual consistency) [2]. The three main desirable properties for this paradigm are:

a) Consistency - The Read operation confirms all previously completed Write operations.

b) Available - Guarantee that every query gets a response, whether it succeeds or fails.

c) Partition tolerance - Guaranteed properties are maintained even if problems occur with the network infrastructure and communication between machines is impossible.

Big data analytics provides valuable guidance and insights that can change the way companies make decisions.

Integrating Big Data and Hadoop into the curriculum provides students with the opportunity to gain hands-on skills in data processing and analysis. By working with real data and using Hadoop, students can learn how to store, process and extract valuable information from datasets. This practical experience prepares them for the future challenges of the job market and makes them competitive candidates.

The interactive nature of Hadoop and its related tools allows students to explore data, create models, and perform real-time analysis. They can conduct experiments, explore trends, and make predictions in learning environments, which improves their critical thinking and problem-solving skills.

III. Features of Apache Hadoop and Methods of Use in Student Training and Education

Apache Hadoop, created 2005, is an open environment for distributed storage and processing of big data. This framework provides scalability, reliability and the ability to process big data in a resource-intensive environment. Hadoop consists of several key components that provide means of administration and monitoring.

The most important advantages of Hadoop for Big Data Management are:

- Scalability: Hadoop can easily scale horizontally, enabling data processing in any multi-server environment.
- Reliability: With HDFS, data is partitioned and distributed across different servers, ensuring that in the event of a crash, the data is still accessible.

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• Data processing: Hadoop MapReduce enables parallel processing of large data on multiple machines, which significantly speeds up analysis time.

• Flexibility: Hadoop is open source and supports a variety of programming languages, allowing developers to use their preferred tools.

Hadoop is used in various fields:

• Business Analytics: Organizations use Hadoop to analyze large volumes of data and discover trends and patterns that support business decisions.

• Scientific research: In fields such as genomics, particle physics, and climate research, Hadoop helps process big scientific data.

• Internet of Things (IoT): Hadoop can be used to process data from IoT sensors and devices.

The basic architecture of a Hadoop cluster is shown in Fig. 1. Some of the ***key components in the Hadoop ecosystem are***:

• Client Machines: These are the computers from which users submit queries and tasks to the Hadoop cluster. They use tools like Hue, the command line, or other software to interact with Hadoop.

• Name Node (HDFS): This component manages metadata for the file system (HDFS), keeps track of the location of data blocks, and controls data replication. It also defines where the data blocks are stored.

It is important to note that the Name Node is a critical element for the stability of HDFS. If the Name Node crashes, the data storage and access system may be affected. Because of this, techniques and strategies for fault tolerance and Name Node redundancy are commonly used.

• Data Nodes (HDFS): These servers store the actual data on the Hadoop Distributed File System (HDFS). They replicate the data that is stored in the system to ensure fault tolerance.

• Task Nodes (MapReduce): These servers perform the data processing tasks in the MapReduce virtual environment. They handle the "Map" and "Reduce" tasks that enable parallel processing of the data.

Hadoop key tools are:

• Hadoop Distributed File System (HDFS): HDFS is the heart of the Hadoop ecosystem and provides distributed data storage. This file system divides the data into blocks and distributes them across multiple servers, which ensures fault tolerance and the possibility of parallel processing.

• MapReduce: MapReduce is a model for processing big data in a distributed environment. It allows developers to define map and reduce operations on data and thus perform parallel processing and analysis.

• YARN (Yet Another Resource Negotiator): YARN is a resource management system that enables dynamic management and allocation of resources in the cluster. This allows multiple applications to run on a single Hadoop environment.

• Apache Hive: Hive is a data analytics infrastructure layer on top of Hadoop. It provides a SQL-like language (HiveQL) that allows analysts to extract data from Hadoop without the need to program.

• Apache Pig: Pig provides a high-level scripting language called Pig Latin for data processing in Hadoop. This language allows users to define complex data operations to be performed on Hadoop.

• Apache HBase: HBase is a non-relational database that provides fast access to large volumes of structured data. It is built on top of HDFS and is suitable for applications that require low read and write latency.

• Apache Spark: Although not part of the core Hadoop project, Spark is an extremely popular tool for real-time data processing and analytics. It provides in-memory data processing capabilities and supports multiple APIs for Python, Scala, Java, and R.

• Apache ZooKeeper: ZooKeeper is a coordination service that is used to manage distributed applications and services in the Hadoop environment.

• Apache Sqoop: Sqoop is a data transfer tool between Hadoop and relational databases.

• Apache Flume: Flume is a tool for collecting, aggregating and transporting data from various sources to Hadoop.

On the basis of the combination of Hadoop and Hive, large projects have been implemented in various areas:

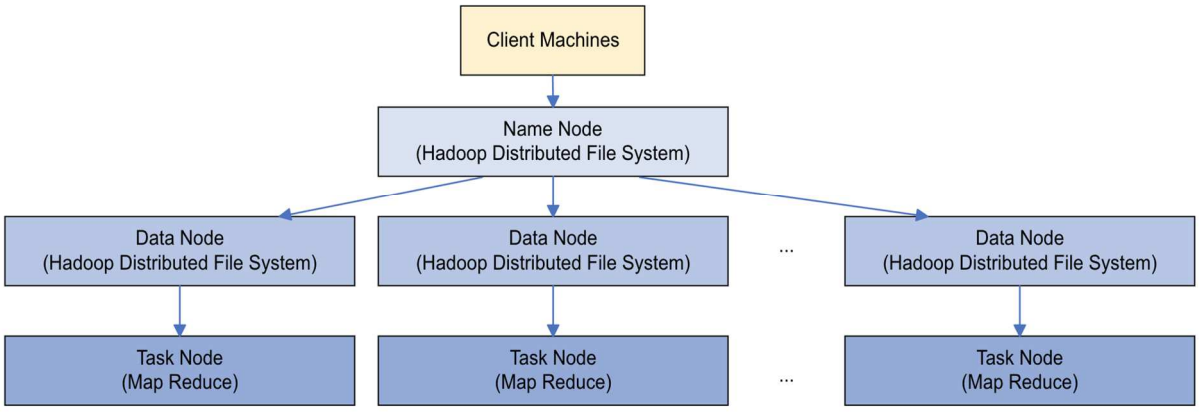

Fig. 1. Hadoop architecture

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Social Networks - [3] Hive is a warehouse of Facebook data in 800,000 tables, generating petabytes of new data every day, executing over 600,000 queries, and having over a million mapreduce jobs active. In [4] the authors describe the use of a combination of Hadoop and Hive for analyzing Twitter data, and in [5] the capabilities of Hadoop for analyzing YouTube comments.

Medicine - For processing large volumes of data about brain damage for the purpose of analyzing survival predictions of patients with brain injuries [6]. Implementations of various intelligent medical systems for processing big data with Hadoop are described in [7, 8] in combination with technologies such as blockchain to increase security [9] or to process medical images [10].

Climate - [11] describes a system that uses historical weather data of a given region and applies MapReduce and Hadoop techniques to analyze this historical data. The development of IoT devices, sensors and methods for collecting and transmitting data from them leads to the accumulation of large volumes of data on climate change, which need to be quickly and efficiently processed and analyzed. Solutions for such Hadoop-based systems are described in [12, 13].

Energy - The efficient use of energy sources and energy consumption is a very current topic in scientific research in recent years. Careful and reasonable use requires analysis of production data and consumption data. In this direction, there are not a few examples of research and systems based on Hadoop and its components [14, 15, 16].

Telecommunications - Telecommunications companies deal with terabytes to petabytes of data every day. IoT devices and related applications in telecommunications further contribute to the growth of this data flow. Recent advances in the field of big data analytics have revealed new possibilities for the organization, storage, analysis and practical use of information from telecoms' big data. Taking advantage of the rapidly changing technological landscape of big data analytical tools is important to improve service offerings and customer service. [17, 18, 19].

The standard way to use Hadoop's data analytics tools is by writing code on the command line or using a high-level programming language. For the purposes of teaching students, especially beginners in the use of this environment, both methods are not particularly suitable. Therefore, Hue (Hadoop User Experience) can be used in the learning process, and not only. It is a web-based application that provides a graphical interface for managing and analyzing data on top of Hadoop. It's an intuitive and user-friendly environment that makes it easy to work with Hadoop and its ecosystem, including Apache Hive and other ecosystem editors and tools. Hue provides graphical tools and interfaces that enable users to use and administer Hadoop and related tools without the need for command-line programming. In Fig. 2 shows a view of this environment after a user has logged in and the Hive editor has been selected for working with.

Hue and Hadoop are excellent tools for teaching students about big data and distributed data processing. Using these tools can provide students with real-world experience working with large volumes of data and related technologies. Here are a few ways that the combination of Hue and Hadoop can be used for student learning that we use:

a) Learning Projects

These may include creating tables in Hive, loading data, running analytical queries, and visualizing the results in Hue. This will teach them how to use the data analysis tools in a real scenario. To have enough data in the tables, free suitable datasets can be used, and we personally use data collected from IoT devices and sensors that are collected for our other scientific research projects.

b) Real Data Analysis

We provide students with real data or data simulating a real scenario for them to analyze using Hue and Hadoop. This may include, for example, the analysis of data from social networks (datasets provided for free), sensor devices (in our case we use soil condition or weather data collected by our devices [20, 21]) or other sources (it is not difficult to find suitable datasets).

c) Creating Practice Exercises and workshops

Hands-on exercises where students have to solve specific problems or perform specific tasks using Hue and Hadoop. For example, they may need to analyze data and answer certain questions or create reports and visualizations of the data.

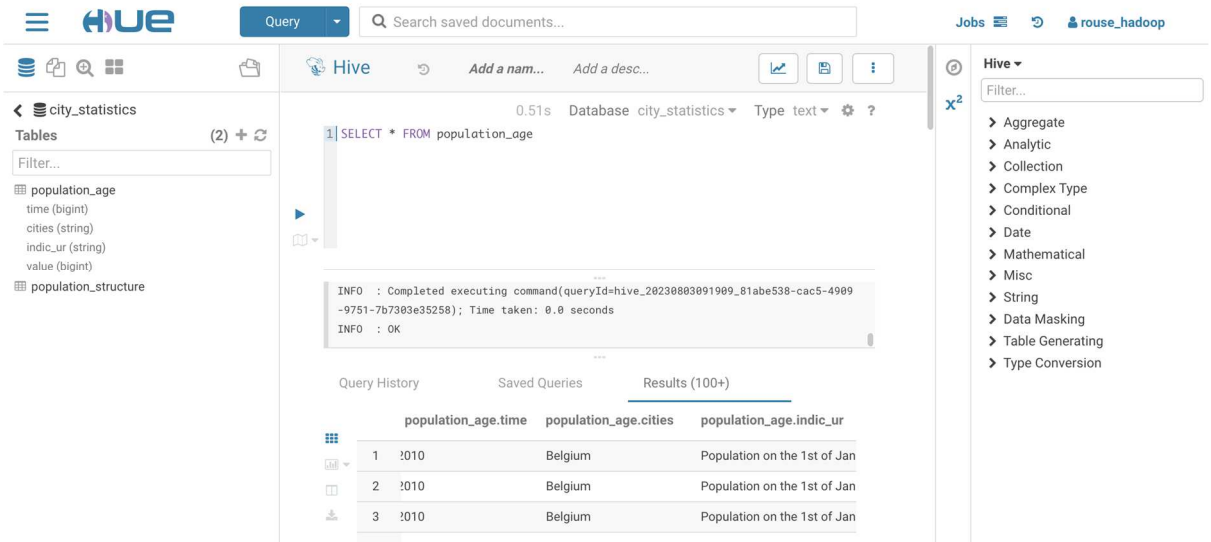

Fig. 2. Hadoop User Experience (HUE) user interface example.

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d) Research Projects

More advanced undergraduates and/or PhD students work on research projects that involve more complex data analysis and processing in Hue and Hadoop. They may be involved in proprietary data model development, query optimization, and more. Some of our master degree students worked in a team on a project to scrape business data on the Internet and analyze it for e-commerce purposes in a Hadoop environment.

e) Integration with Other Tools

Students can learn how to integrate Hue and Hadoop with other tools and technologies, such as Apache Spark, Python, R, and more.

Using Hue and Hadoop to teach students can provide valuable experience and knowledge that is valuable in the job market and business environment where big data plays an increasingly important role.

In the learning process with the students, we mainly use Hive or Impala. In the Hue interface, there is a drop-down menu with various tools and applications (fig.2 the Query drop-down menu). The editor to work with is selected, let's assume Hive. What is the difference between Hive and standard relational database management systems (DBMS) is described in Table 1. It is the awareness of these differences in the process of solving practical problems that helps students understand the differences between a database and a data warehouse.

TABLE I. DIFFERENCES BETWEEN HIVE AND RELATIONAL DBMS

	Hive	Relational DBMS
Schema	Enforces schema on reading	Enforces schema on write
Size of data	petabytes	terabytes
Notion	Write once read many times	Write and read many times
Idea	Data warehouse, supports SQL, but it is not exactly database	Database management systems based on relational data model
Scalability	Easy at low cost	Not scalable at low cost

In Hue, you can create and manage your Hive tables using a graphical interface and follow the steps and instructions in the dialog, and you can use Hive's SQL-like language (HiveQL) (fig. 3). You can also load data into these tables by uploading files from your device or by writing appropriate HiveQL query commands.

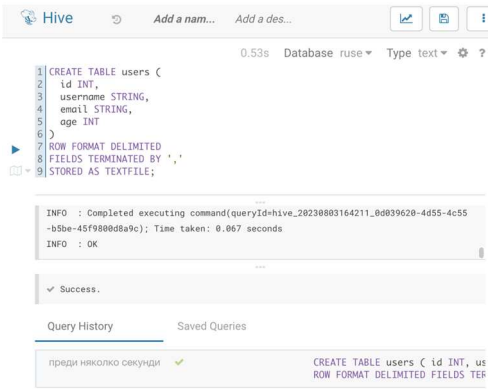

Fig. 3. HiveQL for creating table in Hive.

Hue has a graphical query editor where HiveQL queries can be written and executed (Fig.4). The familiar and generally popular SQL syntax for students can be used, while Hue converts the queries into a format understood by Hadoop.

Hue provides tools to visualize query results in tabular and graphical form. Various types of graphs, charts and tables can be created to be very helpful in analyzing the data used (fig.5).

Fig. 4. HiveQL for loading data in table.

Using Hadoop with Hue in the learning process with students can provide multiple benefits and enrich their experience in big data processing and data analysis. Here are some of the main benefits:

- Real Scenario of Working with Data: Hadoop and Hue simulate the real process of processing and analyzing big data, which allows students to familiarize themselves with the real work environment and the challenges they face in the industry.
- Hands-on Big Data Experience: Students have the opportunity to work with real Big Data, which can enhance their understanding of how such large volumes of data are managed, processed and analyzed.
- Real-Time Learning: Students can use Hue to run Hive queries interactively, observing the results in real-time. This allows them to test and experiment with queries and see immediate results.
- Visual Programming: Hue provides a visual interface for working with data, making programming and using tools like Hive more accessible to beginning students.
- Collaboration and Group Learning: Hadoop with Hue allows students to collaborate and work together in one central cluster, sharing data and resources. This is especially useful for group projects and learning.
- Career Preparation: Working with Hadoop and Hue can prepare students for future careers in fields such as big data processing, data analytics, machine learning, and other fields that require competencies in working with large volumes of data.

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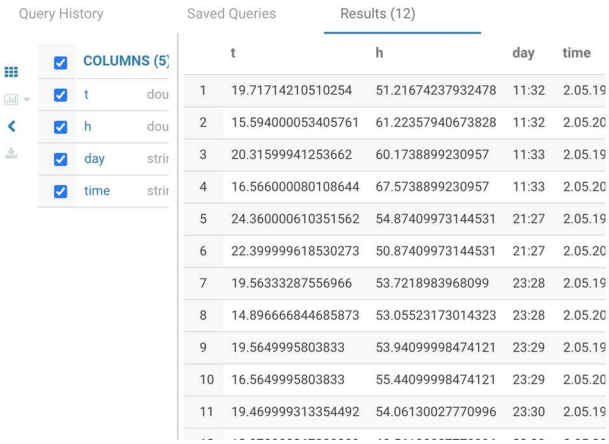
- Variety of Projects and Research: Students can build and run a variety of projects and research using Hadoop with Hue. This may include data analysis, forecasting, experimental data processing, and more.

- Improving Analytical Skills: Working with big data and tools like Hive can help students develop analytical skills, logical thinking and problem-solving abilities.

- Additional Learning Opportunities: Hadoop with Hue offers a wide variety of tools and technologies in the Hadoop ecosystem that students can explore and learn.



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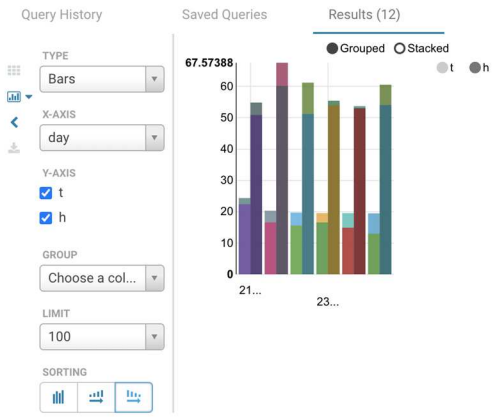

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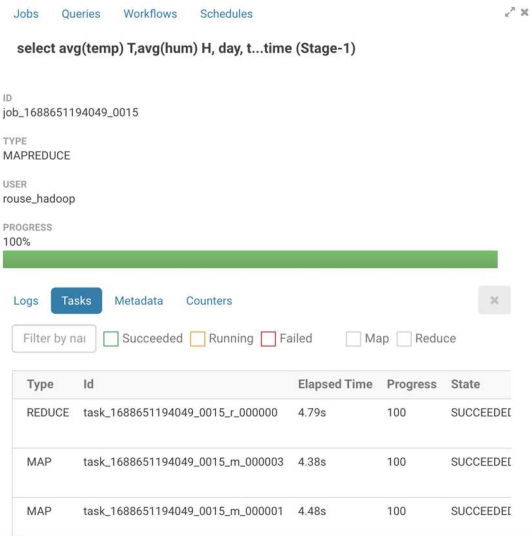

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Fig. 5. HiveQL execution results a) in text/tabular form; b) graphic interpretation; c) description of the query execution process.

IV. Conclusions

Apache Hadoop is a powerful tool for managing and analyzing big data. This framework enables organizations to address the challenges of processing large volumes of information and extract valuable knowledge from it. With its scalability, reliability and parallel processing capabilities, Hadoop continues to be a key technology in the field of big data.

Using Hadoop with Hue in the learning process can enrich students' learning experiences by providing them with practical skills that are valuable for their future careers. This experience can prepare them for the challenges and opportunities that the field of big data processing and data analytics offers.

The combination of Hadoop and Hue can give students the opportunity to become familiar with modern technologies that are widely used in industry and develop skills that are in demand in the job market. They can learn how to process, analyze and visualize large volumes of data, which is critical in the digital age in which we live.

In addition, the use of real data and tasks can make learning more engaging and practical for students, motivating them to develop their own projects, research and solutions. This can encourage them to show creativity and innovative thinking, which is important for their successful careers.

In short, using Hadoop with Hue can enhance the educational process, expand students' practical skills, and prepare them for their future professional challenges in the field of big data processing and data analysis.

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